For Lab 4 I first worked out g, h and f by hand, and S, A and C all came out with f equal to 5, then G came out with 6, so A* gave S, A, C, G for 6, and I checked that every h value was below the real remaining cost, so the heuristic was admissible. Then I ran the same code with all h values set to zero for the UCS test, and it gave the same path and cost but expanded 4 nodes, while A* expanded only 3, because UCS also had to look at B. At the end I changed only h of C from 2 to 10, and A* gave S, B, G for 9 instead, and I first thought the program had a bug, but C really needs only 3, so an overestimate can break the guarantee even if the code is right.